NTUST Laboratory of Digital Logic Design
Project 6
Verilog HDL

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Source Code:

Moore FSM:

```
`timescale 1ns/100ps

module MooreFSM(output y_out,input x_in,clk,rst);

reg [1:0]state,next_state;
parameter a=1'b0,b=1'b1,c=1'b1,d=1'b0;

always @(posedge clk,negedge rst)
    if(rst==0)
        state<=a;
    else
        case(state)
            a: if(~x_in) state<= b; else state<=c;
            b: if(~x_in) state<= c; else state<=d;
            c: if(~x_in) state<= b; else state<=d;
            d: if(~x_in) state<= c; else state<=a;
        endcase
    assign y_out =state;
endmodule
```

Mealy FSM:

```
`timescale 1ns/100ps

module MealyFSM(output reg y_out,input x_in,clk,rst);

reg [1:0]state,next_state;
parameter a=2'b00,b=2'b01,c=2'b10,d=2'b11;

always @(posedge clk,negedge rst)
    if (rst==0) begin
        state <=a;
        y_out=0;
    end
    else
        state<=next_state;

always @(state,x_in)
    case(state)
        a: if(x_in) next_state=c;
            else next_state=b;
        b: if(x_in) next_state=d;
            else next_state=c;
        c: if(x_in) next_state=d;
            else next_state=b;
        d: if(x_in) next_state=a;
            else next_state=c;
    endcase

always @(state,x_in)
    if(rst==0)
        y_out=0;
    else
        case(state)
            a,d: y_out=~x_in;
            b,c: y_out=x_in;
            default:
                y_out=0;
        endcase

endmodule
```

Test Bench:

Moore FSM:

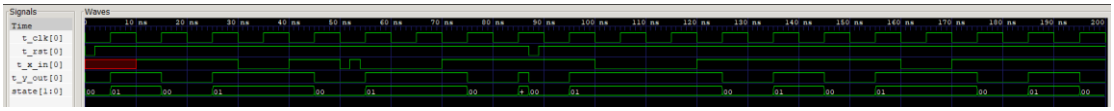
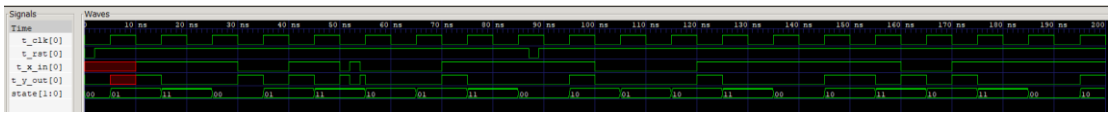
```
`timescale 1ns/100ps
module tMooreBench;
    wire t_y_out;
    reg  t_x_in,t_clk,t_rst;

    MooreFSM M(t_y_out,t_x_in,t_clk,t_rst);
    initial #200 $finish;
    initial begin
        t_clk=0;
        forever #5 t_clk=~t_clk;
    end

    initial fork
        t_rst=0;
     #2  t_rst=1;
     #87 t_rst=0;
     #89 t_rst=1;
     #10 t_x_in=1;
     #30 t_x_in=0;
     #40 t_x_in=1;
     #50 t_x_in=0;
     #52 t_x_in=1;
     #54 t_x_in=0;
     #70 t_x_in=1;
     #80 t_x_in=1;
     #90 t_x_in=1;
     #100 t_x_in=0;
     #120 t_x_in=1;
     #160 t_x_in=0;
     #170 t_x_in=1;
     join
    initial $dumpvars;
endmodule
```

Mealy FSM:

```
`timescale 1ns/100ps

module tMealyBench;
    wire t_y_out;
    reg  t_x_in,t_clk,t_rst;

    MealyFSM M(t_y_out,t_x_in,t_clk,t_rst);
    initial #200 $finish;
    initial begin
        t_clk=0;
        forever #5 t_clk=~t_clk;
    end

    initial fork
        t_rst=0;
     #2  t_rst=1;
     #87 t_rst=0;
     #89 t_rst=1;
     #10 t_x_in=1;
     #30 t_x_in=0;
     #40 t_x_in=1;
     #50 t_x_in=0;
     #52 t_x_in=1;
     #54 t_x_in=0;
     #70 t_x_in=1;
     #80 t_x_in=1;
     #90 t_x_in=1;
     #100 t_x_in=0;
     #120 t_x_in=1;
     #160 t_x_in=0;
     #170 t_x_in=1;
     join
    initial $dumpvars;
endmodule
```

Result:

Moore FSM:



Mealy FSM: